

Technical Document #340: Database Systems - Comprehensive Overview

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Relational databases organize data into tables with rows and columns. SQL is used to query and manipulate structured data.

Sharding distributes data across multiple database servers. It improves performance by parallelizing query processing.

Database monitoring tracks performance metrics and identifies bottlenecks. Query analysis and indexing strategies optimize database performance.

Database migration tools manage schema changes and data transformations. Flyway and Liquibase are popular database migration tools.

NoSQL databases handle unstructured and semi-structured data. Document stores, key-value stores, and graph databases are common NoSQL types.

Database connection pooling reuses database connections to reduce overhead. It improves performance for applications with many concurrent users.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Key Takeaways:

- Database replication copies data across multiple servers.
- ACID properties (Atomicity, Consistency, Isolation, Durability) ensure reliable transaction processing in databases.
- Database indexing improves query performance by creating data structures that allow faster data retrieval.